Electron photon histories with 250 ns buckets

NAUSICAA production analysis

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Run and detector settings

The detector-stage electron branch was replayed from the saved common primary-pair realisation. The two samples use the same source events, placement, pair draws, and attachment streams as the earlier 2 μs run. The ion histories were retained. Pure Xe and Xe–15% He are alternative gas models for each event. The HDF5 inputs to the SiPM response are VUV photons incident on each SiPM per time bucket, before PDE or electronics.

Quantity	Value
Electron time bucket	0.25 μs
Longitudinal bucket width	0.25 mm
Electron drift speed	1 mm/μs
Pressure	15 bar
Helium fraction in mixture	15%
EL hole pitch, diameter, depth	10, 5, 5 mm
Reduced EL field	3 kV/cm/bar
Electron lifetime	15 ms
Primary W, Fano F	21.9 eV, 0.15
EL variance ratio J	0.01

Production ledgers

Sample	Gas	Events	Initial pairs	Surviving e^-	Emitted photons	Incident photons
$0\nu\beta\beta$	Pure Xe	8,000	897,306,359	856,854,917	1,998,549,749,864	381,689,031,793
$0\nu\beta\beta$	Xe–15% He	8,000	897,306,359	856,854,917	1,965,197,038,264	375,318,926,831
Single electrons	Pure Xe	8,000	897,244,198	856,968,162	1,998,813,669,858	381,740,368,430
Single electrons	Xe–15% He	8,000	897,244,198	856,968,162	1,965,456,724,396	375,369,610,548

Every event replays and checks its primary-pair checksum against the saved realisation. The electron/photon ledger requires surviving electrons no greater than initial pairs and incident photons no greater than emitted photons. Photon groups retain event, gas, SiPM and signed bucket indices.

Timing and optical assumptions

Electrons are diffused directly into the physical 10 mm SiPM grid and 0.25 mm longitudinal buckets. EL yield and its J fluctuation, electron attachment and analytic direct optical acceptance follow the resolved detector card and `docs/production_formulas.tex`. Photon time is assigned to the parent electron's arrival bucket at the EL hole. The model does not resolve electron flight or photon emission time within the 5 mm hole. The finer 250 ns bucket therefore resolves the saved arrival-coordinate model; it does not add a measured EL time profile.

The numerical source and full provenance are stored in the HDF5 summary under the production pilot results directory. Its filename is `electron_250ns_summary_` followed by the run identity `464425eac32a2c34` and `.h5`. Large branch histories remain local beside that summary.